

Module 2

Fine Tuning an LLM

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Fine Tuning an LLM

- When custom data is added to an LLM using RAG, the model itself is not updated
 - All the parameters stay the same
 - Vectors representing the new knowledge are computed and stored in a database
 - The LLM retrieves these vectors (i.e., the specific data chunks) and combines it with its model (the LLM network parameters) to figure out the output
 - The LLM itself is unchanged
- In fine tuning
 - The model is updated with new parameters
 - You get, in essence, a new LLM
- When fine tuning, you need to go through all the steps in the machine learning process

Fine Tuning: Broad Steps

- Gather data: The quality of the data is the most important input into a model. Data should be representative of the domain you are customizing on; should be sufficient in quantity
- Preprocessing and feature engineering
 - Clean the data
 - Create appropriate features
- Split into training/validation/testing sets: Fine tuning changes model parameters and you need to ensure that the model is learning “correctly”
- Fine tune the model
- Test the fine tuned model

Supervised Fine Tuning: Process

Supervised fine tuning

- The pre-trained LLM is given specific labeled examples
 - A prompt
 - *How can I return my LCD TV?*
 - A response
 - *To return your LED TV, ensure it is in its original packaging and includes all accessories. Returns are accepted within 30 days of purchase. A restocking fee of 15% may apply.*
- The prompt is used to generate a response
 - Example: A generic response on returns
- The generated response is compared with the labeled response
- Weights are adjusted to account for the error
- The process is repeated

Supervised Fine Tuning: Advantages and Disadvantages

- Advantages:
 - Lower processing and memory requirements than full retraining
 - Fall back to pre-trained model is more seamless (compared to RAG)
 - Adapts to the specific domain (like RAG)
- Disadvantages:
 - Model weights are changed and this may compromise the reliability of the LLM for non-domain questions
 - Overfitting: This is a big danger since the model weights are changed. General queries may still give a domain specific response even where they are not suitable
 - Data issues: Data has to be reconstituted in a prompt response format and this may not be practical
 - For example, many different forms of the LCD TV prompt need to be prepared
 - And this has to be repeated for all possible prompt/response pairs

Instruction Tuning

- In instruction tuning, the pre-trained LLM is provided with an instruction and given a response
 - Instruction: Translate *I love you* into Italian
 - Response: te amo
- Typical use cases:
 - Language translation
 - Multiple choice tests
- Instruction fine tuning works like supervised fine tuning (the model is updated) but is used
 - when there is insufficient labeled data
 - When the response is well defined

Parameter Efficient Fine Tuning (PEFT)

- Supervised fine tuning and instruction fine tuning update the entire model
- Since an LLM is huge (trillions of parameters), these methods are relatively resource intensive
 - Though less intensive than a complete retraining!
- Parameter efficient fine tuning focuses on updating a small part (subset) of the model
 - Can work with less data (since the training is focused)
 - More efficient (since only a small subset of the model is being changed)
 - Less likely to be overfitted (since the LLM is largely unchanged)
 - However, not as reliable (since only a small subset of the LLM is retrained)
- PEFT is mostly used when
 - Resources are limited
 - Data availability is limited

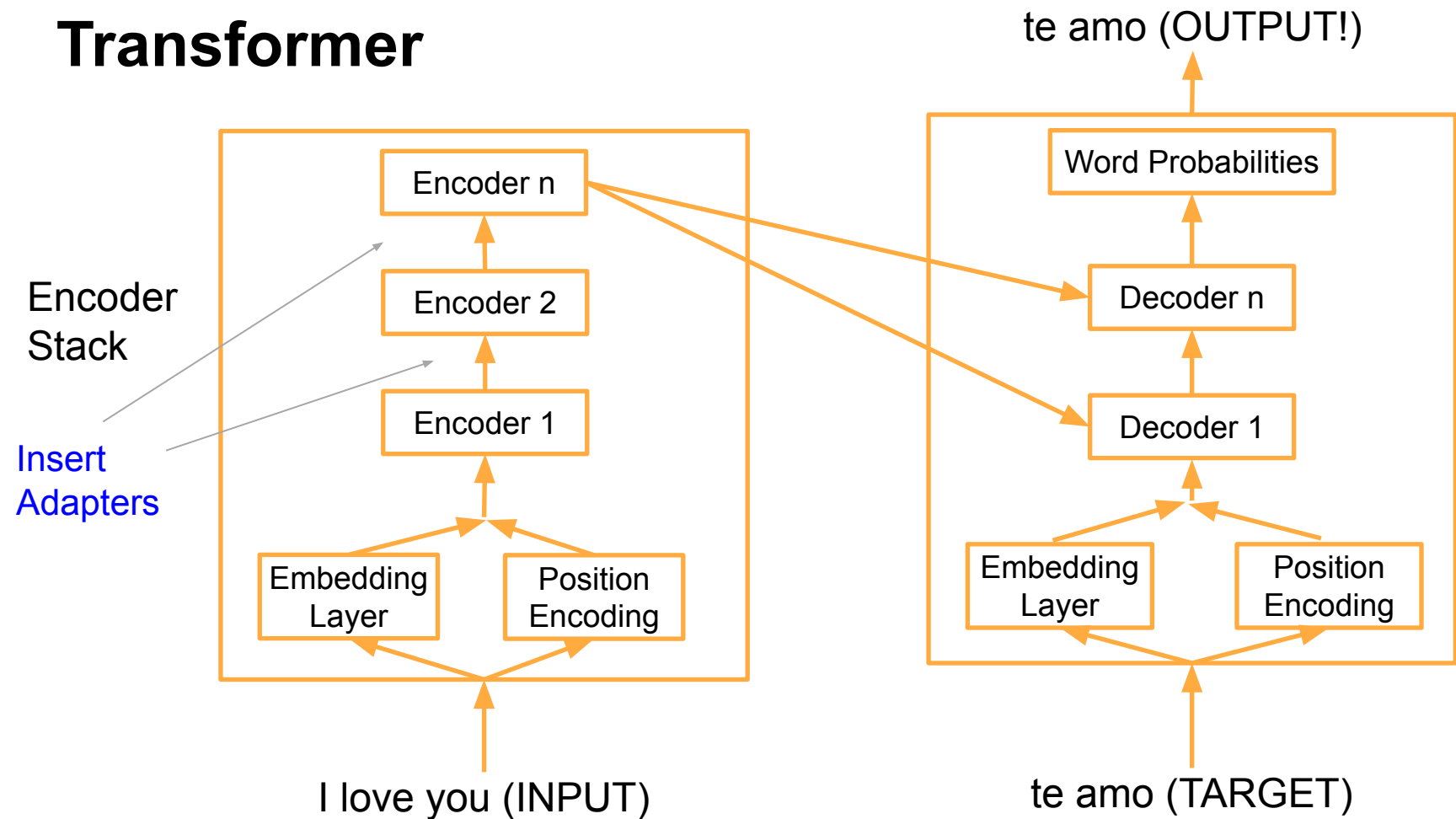
Types of Parameter Efficient Fine Tuning

- Adapters
- Low rank adaption (LORA) and Quantized Low Rank Adaption (QLORA)
- Infused Adapter by Inhibiting and Amplifying Inner Activations (IA3)
- Layer freezing
- Prefix tuning
- Prompt tuning

Adapters

- Adapters are new submodules that are inserted into the transformer architecture
- With each training case (labeled) only the weights in the adapter modules are updated
- The original pre-trained LLM weights are not changed
- Since the adapters are relatively small (few weights) the resources required are relatively low

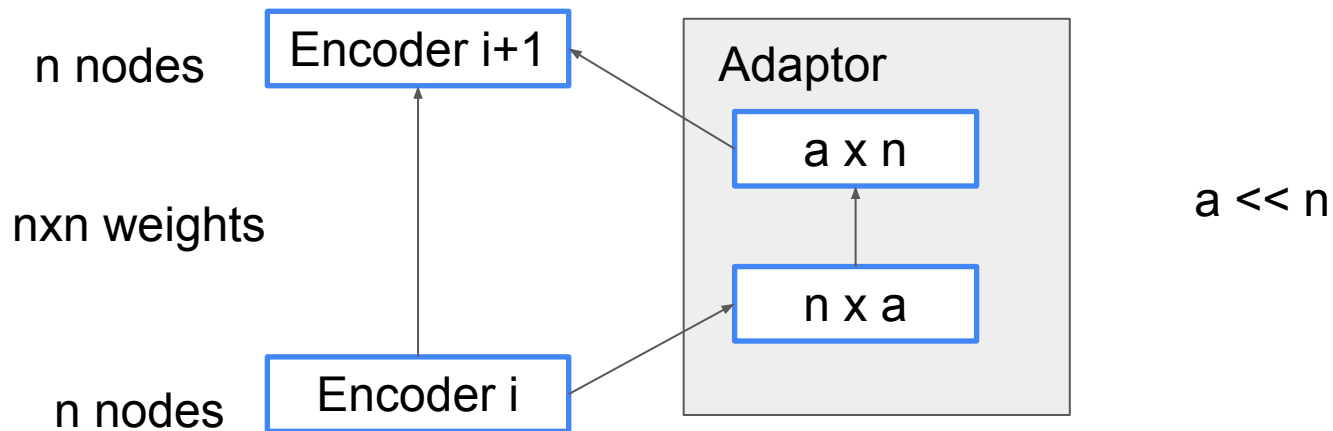
Transformer



LORA: Low Rank Adaptation

- In between any two layers of a transformer, there are $n \times n$ weights
- LORA keeps two smaller matrices
 - $n \times a$ and $a \times n$, where $a \ll n$
- As each case is passed through the transformer
 - The change in weights is computed
 - A lower dimension approximation of this change is used to update the weights in the LORA adaptor
 - The original weights are unchanged
- The main advantage is the reduced memory and processing requirement
 - The LORA adaptor is many (many!) orders of magnitude smaller than the pre-trained LLM

LORA Adaptor



IA3

- IA3 is structurally similar to LORA
- But, the low rank vectors are directly learned rather than computed from the weight changes of the original model
- This makes IA3 faster and more memory efficient than LORA

Layer Freezing

- Roughly
 - The early layers in a model are more general
 - Language elements
 - General knowledge
 - Later layers are more specialized
 - Domain specific knowledge
 - Derived knowledge
 - How to knowledge (classify, summarize, translate, etc.)
- Layer freezing attempts to freeze early layers and update weights only in later layers when fine tuning a model
- Models can be trained to figure out which layers (or parameters) need to be updated (beyond the scope of our class!)

Prefix Tuning

- A vector is prepended to the model, before the input
- The purpose of the vector is to provide an operational context to the LLM
- For example:
 - A prefix vector may guide the model to produce a summary of the input
 - A prefix vector may guide the model to produce a translation of the input
 - A prefix vector may guide the model to set the context to Olympics
 - A prefix vector may guide the model to set the context to the presidential elections
- Advantages and disadvantages
 - Very memory efficient (a vector) and fast training (only the prefix vector is updated)
 - Limited use since it is setting a context rather than building new information into the LLM

Prompt Tuning

- The same prompt can be written in many different ways
 - How do I return my LCD TV
 - I bought an LCD TV from your store and now realize it is too big for the space and want to return it. What should I do?
- The above two examples ask the same question but in different ways
- In prompt tuning, a preprocessing layer that is trained with sample prompts is inserted between the input and the model
 - The preprocessing layer adds a set of embeddings to the prompt
 - These embeddings direct the prompt (sort of) toward a standard prompt
 - In the two examples above, both prompts will be directed toward the same question
- No new knowledge is added but the model can be guided toward a specific purpose

Fine Tuning Example